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Roots of polynomial equations

Fundamental Theorem of Algebra

Fundamental Theorem Of Algebra :

For all non-constant polynomials with $a_i \in \mathbb{C}$,

$$a_n z^n + a_{n-1} z^{n-1} + \cdots + a_1 z + a_0 = a_n (z - \alpha_1)(z - \alpha_2) \cdots (z - \alpha_{n-1})(z - \alpha_n),$$

where $\alpha_i \in \mathbb{C}$

Consequence of Fundamental Theorem of Algebra :

For all non - constant polynomials with $a_i \in \mathbb{R}$,

$$p(z) = a_n z^n + a_{n-1} z^{n-1} + \cdots + a_1 z + a_0$$

then $p(z) = 0 \Leftrightarrow p(\bar{z}) = 0$.

Roots of a quadratic equation

If α and β are the roots of a quadratic equation, $f(x) \equiv ax^2 + bx + c = 0$, then the equation must be of the form

$$f(x) = k(x - \alpha)(x - \beta), \quad k \text{ constant}$$

So
$$k(x - \alpha)(x - \beta) \equiv ax^2 + bx + c$$
$$\Rightarrow k(x^2 - (\alpha + \beta)x + \alpha\beta) \equiv ax^2 + bx + c$$

Equating coefficients yields

$$k = a$$

$$-k(\alpha + \beta) = b$$

$$k\alpha\beta = c$$

Therefore we obtain

$$\alpha + \beta = -\frac{b}{a}, \quad \alpha\beta = \frac{c}{a}.$$

Find similar relations for cubics and quartics.

We can now think of the quadratic equation as

$$x^2 - (\text{sum of roots})x + (\text{product of roots}) = 0.$$

Quadratic, cubic, quartic, quintic

Quadratic	Cubic	Quartic	Quintic
$\sum \alpha = -\frac{b}{a}$	$\sum \alpha = -\frac{b}{a}$	$\sum \alpha = -\frac{b}{a}$	$\sum \alpha = -\frac{b}{a}$
$\alpha\beta = \frac{c}{a}$	$\sum \alpha\beta = \frac{c}{a}$	$\sum \alpha\beta = \frac{c}{a}$	$\sum \alpha\beta = \frac{c}{a}$
	$\alpha\beta\gamma = -\frac{d}{a}$	$\sum \alpha\beta\gamma = -\frac{d}{a}$	$\sum \alpha\beta\gamma = -\frac{d}{a}$
		$\alpha\beta\gamma\delta = \frac{e}{a}$	$\sum \alpha\beta\gamma\delta = \frac{e}{a}$
			$\alpha\beta\gamma\delta\epsilon = -\frac{f}{a}$

WORKED EXAMPLE 1.1

The quadratic equation $x^2 - 2px + p = 0$ is such that one root is three times the value of the other root. Find p .

Answer

$$\alpha + 3\alpha = 2p$$

$$p = 2\alpha$$

Using $\alpha + \beta = -\frac{b}{a}$.

$$\alpha \times 3\alpha = p$$

$$p = 3\alpha^2$$

Using $\alpha\beta = \frac{c}{a}$.

$$\frac{p}{3} = \left(\frac{p}{2}\right)^2$$

Equate the two results.

$$4p - 3p^2 = 0$$

Cross multiply.

$$p = \frac{4}{3}$$

Factorise and omit the case when $p = 0$.

S_1, S_2 and S_{-1} for quadratic roots

Let $S_n = \alpha^n + \beta^n$. Then $S_1 = \alpha + \beta = \sum \alpha$.

Since $(\alpha + \beta)^2 = \alpha^2 + 2\alpha\beta + \beta^2$ we have

$$S_2 = \sum \alpha^2 = \alpha^2 + \beta^2 = (\alpha + \beta)^2 - 2\alpha\beta = (\sum \alpha)^2 - 2 \sum \alpha\beta$$

Since $\frac{1}{\alpha} + \frac{1}{\beta} = \frac{\alpha+\beta}{\alpha\beta}$ we have

$$S_{-1} = \sum \frac{1}{\alpha} = \frac{1}{\alpha} + \frac{1}{\beta} = \frac{\alpha+\beta}{\alpha\beta} = \frac{\sum \alpha}{\sum \alpha\beta}$$

What are S_0 and S_{-2} for quadratics?

S_3 for quadratics

WORKED EXAMPLE 1.2

Find $\alpha^3 + \beta^3$ in summation notation.

Answer

$$(\alpha + \beta)^3 = \alpha^3 + 3\alpha^2\beta + 3\alpha\beta^2 + \beta^3$$

Use the binomial expansion for $(x + y)^n$.

$$\alpha^3 + \beta^3 = (\alpha + \beta)^3 - 3\alpha\beta(\alpha + \beta)$$

Rearrange and factorise.

$$\Sigma\alpha^3 = (\Sigma\alpha)^3 - 3\Sigma\alpha\beta\Sigma\alpha$$

Sum for each possible root.

Alternatively, use $-3\Sigma\alpha^2\beta$ in place of $-3\Sigma\alpha\beta\Sigma\alpha$. However, it is not as easy to calculate with this form.

$$S_3 = \sum \alpha^3 = \alpha^3 + \beta^3 = \left(\sum \alpha \right)^3 - 3 \sum \alpha\beta \sum \alpha$$

WORKED EXAMPLE 1.3

Given that $2x^2 + 3x - 2 = 0$ has roots α, β , find the values of $\alpha^2 + \beta^2$ and $\alpha^3 + \beta^3$.

Answer

$$2\alpha^2 + 3\alpha - 2 = 0$$

$$2\beta^2 + 3\beta - 2 = 0$$

$$\Rightarrow 2S_2 + 3S_1 - 4 = 0$$

Add the two equations to get the recurrence form.

$$S_1 = -\frac{3}{2}$$

State $S_1 = -\frac{b}{a}$ from the original quadratic equation.

$$S_2 = \alpha^2 + \beta^2 = \frac{17}{4}$$

Substitute the S_1 value into the equation.

$$2x^2 + 3x - 2 = 0$$

$$\Rightarrow 2x^3 + 3x^2 - 2x = 0$$

Multiply by x .

$$\Rightarrow 2S_3 + 3S_2 - 2S_1 = 0$$

Add $2\alpha^3 + 3\alpha^2 - 2\alpha = 0$ and $2\beta^3 + 3\beta^2 - 2\beta = 0$.

$$S_3 = \alpha^3 + \beta^3 = -\frac{63}{8}$$

Use the values of S_1 and S_2 .

EXERCISE 1A

- 1 Each of the following quadratic equations has roots α, β . Find the values of $\alpha + \beta$ and $\alpha\beta$.
a $x^2 + 5x + 9 = 0$ b $x^2 - 4x + 8 = 0$ c $2x^2 + 3x - 7 = 0$
- 2 Given that $3x^2 + 4x + 12 = 0$ has roots α, β , find:
a $\alpha + \beta$ and $\alpha\beta$ b $\alpha^2 + \beta^2$
- 3 $x^2 - (2 + p)x + (7 + p) = 0$ has roots that differ by 1. Find the value of p given that $p > 0$.
- PS** 4 If $a + b = -3$ and $a^2 + b^2 = 7$, find the value of ab and, hence, write down a quadratic equation with roots a and b .
- P** 5 If $x^2 + bx + c = 0$ has roots α and β , prove that:
a if $\alpha = 3\beta$, then $b^2 = \frac{16}{3}c$
b if $\alpha = \beta - 2$, then $b^2 = 4(c + 1)$.
- PS** 6 You are given the quadratic equation $px^2 + qx - 16 = 0$, which has roots α and β . Given also that $\alpha + \beta = -\frac{1}{2}$ and $\alpha\beta = -8$, find the values of p and q .
- 7 The quadratic equation $x^2 + 2x - 6 = 0$ has roots α and β . Find the values of $(\alpha - \beta)^2$ and $\frac{1}{\alpha^2} + \frac{1}{\beta^2}$.
- PS** 8 A quadratic equation has roots α and β . Given that $\frac{1}{\alpha} + \frac{1}{\beta} = \frac{1}{2}$ and $\alpha^2 + \beta^2 = 12$, find two possible quadratic equations that satisfy these values.
- 9 The quadratic equation $3x^2 + 2x - 4 = 0$ has roots α and β . Find the values of S_1, S_2 and S_{-1} .
- PS** 10 You are given the quadratic equation $4x^2 - x + 6 = 0$ which has roots α and β .
a Find $\alpha^2 + \beta^2$.
b Without solving the quadratic equation, state what your value for part **a** tells you about the roots.

Cubics

$$ax^3 + bx^2 + cx + d = 0$$

In[5]:= **Solve**[**a x + b == 0, x]**

Out[5]= $\left\{\left\{x \rightarrow -\frac{b}{a}\right\}\right\}$

In[6]:= **Solve**[**a x^2 + b x + c == 0, x]**

Out[6]= $\left\{\left\{x \rightarrow \frac{-b - \sqrt{b^2 - 4 a c}}{2 a}\right\}, \left\{x \rightarrow \frac{-b + \sqrt{b^2 - 4 a c}}{2 a}\right\}\right\}$

$$az^2 + bz + c = 0 \Rightarrow z = \begin{cases} \frac{-b}{2a} & \text{if } b^2 - 4ac = 0 \\ \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} & \text{if } b^2 - 4ac > 0 \\ \frac{-b \pm i\sqrt{4ac - b^2}}{2a} & \text{if } b^2 - 4ac < 0 \end{cases}$$

In[5]:= **Solve**[a x + b == 0, x]

Out[5]= $\left\{\left\{x \rightarrow -\frac{b}{a}\right\}\right\}$

In[6]:= **Solve**[a x^2 + b x + c == 0, x]

Out[6]= $\left\{\left\{x \rightarrow \frac{-b - \sqrt{b^2 - 4 a c}}{2 a}\right\}, \left\{x \rightarrow \frac{-b + \sqrt{b^2 - 4 a c}}{2 a}\right\}\right\}$

In[7]:= **Solve**[a x^3 + b x^2 + c x + d == 0, x]

Out[7]=
$$\left\{\left\{x \rightarrow -\frac{b}{3 a} - \frac{2^{1/3} \left(-b^2 + 3 a c\right)}{3 a \left(-2 b^3 + 9 a b c - 27 a^2 d + \sqrt{4 \left(-b^2 + 3 a c\right)^3 + \left(-2 b^3 + 9 a b c - 27 a^2 d\right)^2}\right)^{1/3}} + \frac{\left(-2 b^3 + 9 a b c - 27 a^2 d + \sqrt{4 \left(-b^2 + 3 a c\right)^3 + \left(-2 b^3 + 9 a b c - 27 a^2 d\right)^2}\right)^{1/3}}{3 \times 2^{1/3} a}\right\}, \left\{x \rightarrow -\frac{b}{3 a} + \frac{\left(1 + i \sqrt{3}\right) \left(-b^2 + 3 a c\right)}{3 \times 2^{2/3} a \left(-2 b^3 + 9 a b c - 27 a^2 d + \sqrt{4 \left(-b^2 + 3 a c\right)^3 + \left(-2 b^3 + 9 a b c - 27 a^2 d\right)^2}\right)^{1/3}} - \frac{\left(1 - i \sqrt{3}\right) \left(-2 b^3 + 9 a b c - 27 a^2 d + \sqrt{4 \left(-b^2 + 3 a c\right)^3 + \left(-2 b^3 + 9 a b c - 27 a^2 d\right)^2}\right)^{1/3}}{6 \times 2^{1/3} a}\right\}, \left\{x \rightarrow -\frac{b}{3 a} + \frac{\left(1 - i \sqrt{3}\right) \left(-b^2 + 3 a c\right)}{3 \times 2^{2/3} a \left(-2 b^3 + 9 a b c - 27 a^2 d + \sqrt{4 \left(-b^2 + 3 a c\right)^3 + \left(-2 b^3 + 9 a b c - 27 a^2 d\right)^2}\right)^{1/3}} - \frac{\left(1 + i \sqrt{3}\right) \left(-2 b^3 + 9 a b c - 27 a^2 d + \sqrt{4 \left(-b^2 + 3 a c\right)^3 + \left(-2 b^3 + 9 a b c - 27 a^2 d\right)^2}\right)^{1/3}}{6 \times 2^{1/3} a}\right\}\right\}$$

Cubics

$$ax^3 + bx^2 + cx + d = 0$$

$$\alpha + \beta + \gamma = \sum \alpha = -\frac{b}{a}$$

$$\alpha\beta + \alpha\gamma + \beta\gamma = \sum \alpha\beta = \frac{c}{a}$$

$$\alpha\beta\gamma = \sum \alpha\beta\gamma = -\frac{d}{a}$$

S_2, S_{-1} and S_{-2} for cubic roots

$$S_2 = \sum \alpha^2 = \alpha^2 + \beta^2 + \gamma^2 = \left(\sum \alpha \right)^2 - 2 \sum \alpha\beta$$

WORKED EXAMPLE 1.4

Find the summation form for the results $\frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma}$ and $\frac{1}{\alpha^2} + \frac{1}{\beta^2} + \frac{1}{\gamma^2}$.

Answer

$$\frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma} = \frac{\alpha\beta + \alpha\gamma + \beta\gamma}{\alpha\beta\gamma}$$

Combine the fractions.

$$\Rightarrow \sum \frac{1}{\alpha} = \frac{\Sigma \alpha\beta}{\Sigma \alpha\beta\gamma}$$

State the result.

$$\frac{1}{\alpha^2} + \frac{1}{\beta^2} + \frac{1}{\gamma^2} = \frac{\alpha^2\beta^2 + \alpha^2\gamma^2 + \beta^2\gamma^2}{\alpha^2\beta^2\gamma^2}$$

Combine the fractions, as before.

$$\sum \frac{1}{\alpha^2} = \frac{\Sigma (\alpha\beta)^2}{(\Sigma \alpha\beta\gamma)^2}$$

State the result.

S_3 for cubics

$$S_3 = \sum \alpha^3 = \alpha^3 + \beta^3 + \gamma^3 = \left(\sum \alpha \right)^3 - 3 \sum \alpha\beta \sum \alpha + 3 \sum \alpha\beta\gamma$$

WORKED EXAMPLE 1.5

Given that $x^3 + 2x^2 + 5 = 0$, find, using summation form, the values of S_1 , S_2 , S_3 and S_{-1} .

Answer

$$S_1 = -2$$

Recall this is $-\frac{b}{a} = -\frac{2}{1}$.

$$S_2 = (\sum \alpha)^2 - 2\sum \alpha\beta$$

$$\Rightarrow S_2 = (-2)^2 - 2 \times 0 = 4$$

Recall the value of $\sum \alpha\beta$ is given by $\frac{c}{a} = \frac{0}{1}$ as the linear term coefficient is 0.

$$S_3 = (\sum \alpha)^3 - 3\sum \alpha\beta \sum \alpha + 3\sum \alpha\beta\gamma$$

The last term is $3\alpha\beta\gamma$.

$$S_3 = (-2)^3 - 3 \times (0) \times (-2) + 3 \times (-5)$$

Substitute the values into the equation.

$$S_3 = -23$$

$$S_{-1} = \frac{0}{-5} = 0$$

Recall that this result is equivalent to $\frac{c}{-d}$, which is obtained by taking the negative of the coefficient of the linear term and dividing by the constant term.

A better way: using recurrence relations to find S_n for cubics

$$S_n = \sum \alpha^n = \alpha^n + \beta^n + \gamma^n$$

$$ax^3 + bx^2 + cx + d = 0$$

$$aS_{n+3} + bS_{n+2} + cS_{n+1} + dS_n = 0$$

If $2x^3 + x^2 + x + 1 = 0$ has roots α, β and γ , what is:

(a) $\alpha + \beta + \gamma$?

(b) $\alpha^2 + \beta^2 + \gamma^2$?

(c) $\alpha^3 + \beta^3 + \gamma^3$?

(d) $\frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma}$?

(e) $\frac{1}{\alpha^2} + \frac{1}{\beta^2} + \frac{1}{\gamma^2}$?

(f) $\frac{1}{\alpha^3} + \frac{1}{\beta^3} + \frac{1}{\gamma^3}$?

WORKED EXAMPLE 1.6

For the cubic equation $3x^3 + 2x^2 - 4x + 1 = 0$, find the value of S_3 .

Answer

$$S_1 = -\frac{2}{3}$$

From $-\frac{b}{a}$.

$$S_2 = \left(-\frac{2}{3}\right)^2 - 2 \times \left(-\frac{4}{3}\right) = \frac{28}{9}$$

Using $S_2 = (\Sigma\alpha)^2 - 2\Sigma\alpha\beta$.

$$3S_3 + 2S_2 - 4S_1 + 3 = 0$$

We know that $3\alpha^3 + 2\alpha^2 - 4\alpha + 1 = 0$ and similar equations can be made for β and γ . We add the three equations.

$$S_3 = -\frac{107}{27}$$

Substitute for the final result.

WORKED EXAMPLE 1.7

For the cubic equation $x^3 - 3x^2 + 4 = 0$, find the value of S_{-3} .

Answer

$$S_{-1} = \frac{0}{-4} = 0$$

Recall that $S_{-1} = \frac{\Sigma\alpha\beta}{\Sigma\alpha\beta\gamma}$.

$$x^3 - 3x^2 + 4 = 0$$

$$\Rightarrow 1 - \frac{3}{x} + \frac{4}{x^3} = 0$$

Divide by x^3 .

$$3 - 3S_{-1} + 4S_{-3} = 0$$

Remember that the 1 is counted three times when adding the three equations for α , β and γ .

$$S_{-3} = -\frac{3}{4}$$

Substitute values into the equation.

EXERCISE 1B

1 Each of the following cubic equations has roots α, β, γ . Find, for each case, $\alpha + \beta + \gamma$ and $\alpha\beta\gamma$.

a $x^3 + 3x^2 - 5 = 0$

b $2x^3 + 5x^2 - 6 = 0$

c $x^3 + 7x - 9 = 0$

2 Given that $x^3 - 3x^2 + 12 = 0$ has roots α, β, γ , find the following values:

a $\alpha + \beta + \gamma$ and $\alpha\beta + \alpha\gamma + \beta\gamma$

b $\alpha^2 + \beta^2 + \gamma^2$

3 The roots of each of the following cubic equations are α, β, γ . In each case, find the values of S_2 and S_{-1} .

a $x^3 - 2x^2 + 5 = 0$

b $3x^3 + 4x - 1 = 0$

c $x^3 + 3x^2 + 5x - 7 = 0$

4 The cubic equation $x^3 - x + 7 = 0$ has roots α, β, γ . Find the values of $\Sigma\alpha$ and $\Sigma\alpha^2$.

M

5 Given that $2x^3 + 5x^2 + 1 = 0$ has roots α, β, γ , and that $S_n = \alpha^n + \beta^n + \gamma^n$, find the values of S_2 and S_3 .

M

PS

6 The cubic equation $x^3 + ax^2 + bx + a = 0$ has roots α, β, γ , and the constants a, b are real and positive.

a Find, in terms of a and b , the values of $\Sigma\alpha$ and $\Sigma\frac{1}{\alpha}$.

b Given that $\Sigma\alpha = \Sigma\frac{1}{\alpha}$, does this cubic equation have complex roots? Give a reason for your answer.

M

PS

7 The cubic equation $x^3 - x + 3 = 0$ has roots α, β, γ .

a Using the relation $S_n = \alpha^n + \beta^n + \gamma^n$, or otherwise, find the value of S_4 .

b By considering S_1 and S_4 , determine the value of $\alpha^3(\beta + \gamma) + \beta^3(\alpha + \gamma) + \gamma^3(\alpha + \beta)$.

P

8 A cubic polynomial is given as $2x^3 - x^2 + x - 5 = 0$, having roots α, β, γ .

a Show that $2S_{n+3} - S_{n+2} + S_{n+1} - 5S_n = 0$.

b Find the value of S_{-2} .

M

9 The cubic equation $px^3 + qx^2 + r = 0$ has roots α, β, γ . Find, in terms of p, q, r :

a S_1

b S_2

c S_3

10 The equation $x^3 + px^2 + qx + r = 0$ is such that $S_1 = 0$, $S_2 = -2$ and $S_{-1} = \frac{1}{5}$.

Find the values of the constants p, q, r .

$$z^2 + 2z + 1$$

2 real roots

$$z^3 - 3z^2 + 3z - 1$$

3 real roots

$$z^2 + 5z - 6$$

2 real roots

$$z^3 - 6z^2 + 11z - 6$$

3 real roots

$$z^2 + 5z + 5$$

2 real roots

$$z^3 + z^2 + z + 1$$

1 real root
2 complex conjugate roots

$$z^2 + 1$$

2 complex conjugate roots

$$z^3 + z^2 + z + 2$$

1 real root
2 complex conjugate roots

$$z^2 + z + 1$$

2 complex conjugate roots

$$z^3 + i$$

3 complex roots
none conjugate

$$z^2 - 3(1 + i)z + 5i$$

2 complex roots
not conjugate

$$z^3 + z^2 - 2z - 1$$

3 real roots

Quartics

$$ax^4 + bx^3 + cx^2 + dx + e = 0$$

$$\alpha + \beta + \gamma + \delta = \sum \alpha = -\frac{b}{a}$$

$$\alpha\beta + \alpha\gamma + \alpha\delta + \beta\gamma + \beta\delta + \gamma\delta = \sum \alpha\beta = \frac{c}{a}$$

$$\alpha\beta\gamma + \alpha\beta\delta + \alpha\gamma\delta + \beta\gamma\delta = \sum \alpha\beta\gamma = -\frac{d}{a}$$

$$\alpha\beta\gamma\delta = \sum \alpha\beta\gamma\delta = \frac{e}{a}$$

A better way: using recurrence relations to find S_n for quartics

$$S_n = \sum \alpha^n = \alpha^n + \beta^n + \gamma^n + \delta^n$$

$$ax^4 + bx^3 + cx^2 + dx + e = 0$$

$$aS_{n+4} + bS_{n+3} + cS_{n+2} + dS_{n+1} + eS_n = 0$$

WORKED EXAMPLE 1.8

A quartic polynomial is given as $x^4 + 3x^2 - x + 5 = 0$ and has roots $\alpha, \beta, \gamma, \delta$. Find the values of S_2 and S_4 .

Answer

$$S_1 = 0$$

Simply state the negative of the coefficient of x^3 , as the coefficient of x^4 is 1.

$$S_2 = 0^2 - 2 \times 3 = -6$$

Use $S_2 = (\sum \alpha)^2 - 2\sum \alpha\beta$.

$$S_4 + 3S_2 - S_1 + 20 = 0$$

Use $S_n = \alpha^n + \beta^n + \gamma^n + \delta^n$.

$$S_4 = -2$$

Final answer.

WORKED EXAMPLE 1.9

For the quartic $x^4 - x^3 + 2x^2 - 2x - 5 = 0$, state the values of S_1 and S_{-1} , and determine the value of S_2 . State whether or not there are any complex solutions.

Answer

$$S_1 = 1$$

$$-1 \times (-1)$$

$$S_{-1} = \frac{-(-2)}{-5} = -\frac{2}{5}$$

$$\text{Use } S_{-1} = \frac{\sum \alpha\beta\gamma}{\sum \alpha\beta\gamma\delta}.$$

$$S_2 = (1)^2 - 2 \times 2 = -3$$

$$\text{Use } S_2 = (\sum \alpha)^2 - 2\sum \alpha\beta.$$

$S_2 < 0$ so there are complex solutions.

EXERCISE 1C

- 1 For each of the following quartic equations, find the values of $\Sigma\alpha$ and $\Sigma\alpha\beta$.
 - a $x^4 - 2x^3 + 5x^2 + 7 = 0$
 - b $2x^4 + 5x^3 - 3x + 4 = 0$
 - c $3x^4 - 2x^2 + 9x - 11 = 0$
- 2 The quartic equation $5x^4 - 3x^3 + x - 13 = 0$ has roots $\alpha, \beta, \gamma, \delta$. Find:
 - a $\Sigma\alpha$ and $\Sigma\alpha^2$
 - b $\Sigma\frac{1}{\alpha}$
- 3 A quartic equation is given as $x^4 + x + 2 = 0$. It has roots $\alpha, \beta, \gamma, \delta$. State the values of S_1 and S_{-1} , and find the value of S_2 .
- PS** 4 The quartic equation $2x^4 + x^3 - x + 7 = 0$ has roots $\alpha, \beta, \gamma, \delta$. Given that $S_3 = \frac{11}{8}$, and using S_n , find the value of S_4 .
- 5 You are given that $x^4 - x^3 + x + 2 = 0$, where the roots are $\alpha, \beta, \gamma, \delta$. Find the values of $\Sigma\alpha, \Sigma\alpha^2$ and $\Sigma\frac{1}{\alpha}$. Hence, determine the value of $\Sigma\alpha^3$.
- PS** 6 The quartic polynomial $x^4 + ax^2 + bx + 1 = 0$ has roots $\alpha, \beta, \gamma, \delta$. Given that $S_2 = S_{-1}$, find S_3 in terms of a .
- PS** 7 The polynomial $3x^4 + 2x^3 + 7x^2 + 4 = 0$ has roots $\alpha, \beta, \gamma, \delta$, where $S_n = \alpha^n + \beta^n + \gamma^n + \delta^n$.
 - a Find the values of S_1 and S_2 .
 - b Find the values of S_3 and S_4 .
 - c Are there any complex roots? Give a reason for your answer.
- PS** 8 For the polynomial $x^4 + ax^3 + bx^2 + c = 0$, with roots α, β, γ and δ , it is given that $\alpha + \beta + \gamma + \delta = 2$, $\alpha\beta\gamma\delta = 1$ and $\alpha^2 + \beta^2 + \gamma^2 + \delta^2 = 0$. Find the values of the coefficients a, b and c .
- P** 9 The roots of the quartic $x^4 - 2x^3 + x^2 - 4 = 0$ are $\alpha, \beta, \gamma, \delta$. Show that $S_4 = 9S_3$.

Substitutions (creating new polynomials)

WORKED EXAMPLE 1.10

Given that $x^2 - 2x + 12 = 0$ has roots α, β , find the quadratic equation with roots $\frac{\alpha}{3}, \frac{\beta}{3}$.

Answer

$$y = \frac{x}{3} \Rightarrow x = 3y$$

Rearrange to make x the subject.

$$(3y)^2 - 2(3y) + 12 = 0$$

Substitute for x .

$$3y^2 - 2y + 4 = 0$$

Multiply out terms and simplify.

WORKED EXAMPLE 1.11

Given that $x^3 + x^2 - 5 = 0$ has roots α, β, γ , find the cubic equation with roots $\frac{1}{\alpha-2}, \frac{1}{\beta-2}, \frac{1}{\gamma-2}$.

Answer

$$y = \frac{1}{x-2} \Rightarrow xy - 2y = 1$$

$$x = \frac{1+2y}{y}$$

Rearrange to make x the subject.

$$\left(\frac{1+2y}{y}\right)^3 + \left(\frac{1+2y}{y}\right)^2 - 5 = 0$$

Substitute for x .

$$\frac{1+6y+12y^2+8y^3}{y^3} + \frac{1+4y+4y^2}{y^2} - 5 = 0$$

Expand brackets.

$$7y^3 + 16y^2 + 7y + 1 = 0$$

Multiply by y^3 and simplify.

EXPLORE 1.1

The polynomial $x^3 + x - 3 = 0$ has roots α, β, γ . If $\frac{a\alpha+1}{\alpha-b}, \frac{a\beta+1}{\beta-b}, \frac{a\gamma+1}{\gamma-b}$ are the roots of another cubic, what are the conditions on a and b to ensure that these cubics are the same?

WORKED EXAMPLE 1.12

The polynomial $x^4 + x^3 - x + 12 = 0$ has roots $\alpha, \beta, \gamma, \delta$. Find the polynomial with roots $\alpha^2, \beta^2, \gamma^2, \delta^2$.

Answer

$$y = x^2$$

State the substitution.

$$x^4 + 12 = x - x^3$$

Rearrange so that both sides when squared give even terms.

$$x^8 + 24x^4 + 144 = x^2 - 2x^4 + x^6$$

Square both sides.

$$x^8 - x^6 + 26x^4 - x^2 + 144 = 0$$

Simplify.

$$y^4 - y^3 + 26y^2 - y + 144 = 0$$

Use $x^2 = y$.

WORKED EXAMPLE 1.13

The cubic polynomial $x^3 + 5x^2 + 1 = 0$ has roots α, β, γ . Using the substitution $y = x^3$, or otherwise, find the value of S_6 .

Answer

$$x^3 + 5x^2 + 1 = 0$$

(1)

Label the original equation.

$$x^3 + 1 = -5x^2$$

Since we are using x^3 , ensure all terms are arranged so when both sides are cubed, they produce powers that are multiples of 3.

$$x^9 + 3x^6 + 3x^3 + 1 = -125x^6$$

Cube both sides.

$$\Rightarrow x^9 + 128x^6 + 3x^3 + 1 = 0$$

Simplify.

$$\Rightarrow y^3 + 128y^2 + 3y + 1 = 0$$

(2)

Use $x^3 = y$, label the new equation.

$$S_1 = -128$$

Determine S_1 .

$$S_2 = (-128)^2 - 2 \times 3 = 16\,378 \text{ for (2)}$$

Substitute for S_2 .

$$\text{Hence, for (1), } S_6 = 16\,378$$

State S_6 .

EXERCISE 1D

M 1 The quadratic equation $x^2 + 5x + 3 = 0$ has roots α, β . Find the quadratic equation with roots $3\alpha, 3\beta$.

M 2 The quadratic equation $2x^2 - 4x + 7 = 0$ has roots α, β .

a Find the quadratic equation with roots α^2, β^2 .

b Find the quadratic equation with roots $2\alpha - 3, 2\beta - 3$.

M 3 Given that $3x^2 - 2x + 9 = 0$ has roots α, β , find the quadratic equation with roots $\frac{\alpha+1}{\alpha}, \frac{\beta+1}{\beta}$.

M 4 The quadratic equation $x^2 - 4x + 9 = 0$ has roots α, β . Find the quadratic that has roots $\frac{1}{\alpha}, \frac{1}{\beta}$.

PS 5 Given that $2x^3 - 5x + 1 = 0$ has roots α, β, γ , find the cubic equation with roots $\alpha^2, \beta^2, \gamma^2$. Hence, find the value of S_4 .

P PS 6 The cubic equation $x^3 + 3x^2 - 1 = 0$ has roots α, β, γ . Show that the cubic equation with roots $\frac{\alpha+2}{\alpha}, \frac{\beta+2}{\beta}, \frac{\gamma+2}{\gamma}$ is $y^3 - 3y^2 - 9y + 3 = 0$. Hence, determine the values of:

a
$$\frac{(\alpha+2)(\beta+2)(\gamma+2)}{\alpha\beta\gamma}$$

b
$$\frac{\alpha}{\alpha+2} + \frac{\beta}{\beta+2} + \frac{\gamma}{\gamma+2}$$

P PS 7 A quartic equation, $2x^4 - x^3 - 6 = 0$, has roots $\alpha, \beta, \gamma, \delta$. Show that the quartic equation with roots $\alpha^3, \beta^3, \gamma^3, \delta^3$ is $8y^4 - y^3 - 18y^2 - 108y - 216 = 0$. Hence, find the values of S_6 and S_{-3} .

PS 8 The cubic equation $x^3 - x + 4 = 0$ has roots α, β, γ . Find the cubic equation that has roots $\alpha^2, \beta^2, \gamma^2$. Hence, or otherwise, determine the values of S_6, S_8 and S_{10} .



WORKED PAST PAPER QUESTION

The equation $x^3 + x - 1 = 0$ has roots α, β, γ .

Show that the equation with roots $\alpha^3, \beta^3, \gamma^3$ is $y^3 - 3y^2 + 4y - 1 = 0$.

Hence, find the value of $\alpha^6 + \beta^6 + \gamma^6$.

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Answer

Start with $x^3 - 1 = -x$.

Then rewrite this as $(x^3 - 1)^3 = -x^3$. This gives $x^9 - 3x^6 + 4x^3 - 1 = 0$.

Let $y = x^3$ to give $y^3 - 3y^2 + 4y - 1 = 0$.

Note that $S_n = \alpha^n + \beta^n + \gamma^n$.

S_6 for the original equation is S_2 for the new equation, so $S_2 = 3^2 - 2 \times 4 = 1$.

Hence, $\alpha^6 + \beta^6 + \gamma^6 = 1$.

- 1 The roots of the equation $x^3 + 4x - 1 = 0$ are α, β and γ . Use the substitution $y = \frac{1}{1+x}$ to show that the equation $6y^3 - 7y^2 + 3y - 1 = 0$ has roots $\frac{1}{\alpha+1}, \frac{1}{\beta+1}$ and $\frac{1}{\gamma+1}$.

For the cases $n = 1$ and $n = 2$, find the value of $\frac{1}{(\alpha+1)^n} + \frac{1}{(\beta+1)^n} + \frac{1}{(\gamma+1)^n}$.

Deduce the value of $\frac{1}{(\alpha+1)^3} + \frac{1}{(\beta+1)^3} + \frac{1}{(\gamma+1)^3}$.

Hence show that $\frac{(\beta+1)(\gamma+1)}{(\alpha+1)^2} + \frac{(\gamma+1)(\alpha+1)}{(\beta+1)^2} + \frac{(\alpha+1)(\beta+1)}{(\gamma+1)^2} = \frac{73}{36}$.

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- 2 The roots of the quartic equation $x^4 + 4x^3 + 2x^2 - 4x + 1 = 0$ are α, β, γ and δ .

Find the values of

- i $\alpha + \beta + \gamma + \delta$,
- ii $\alpha^2 + \beta^2 + \gamma^2 + \delta^2$,
- iii $\frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma} + \frac{1}{\delta}$,
- iv $\frac{\alpha}{\beta\gamma\delta} + \frac{\beta}{\alpha\gamma\delta} + \frac{\gamma}{\alpha\beta\delta} + \frac{\delta}{\alpha\beta\gamma}$.

Using the substitution $y = x + 1$, find a quartic equation in y . Solve this quartic equation and hence find the roots of the equation $x^4 + 4x^3 + 2x^2 - 4x + 1 = 0$.

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- 3 The cubic equation $x^3 - x^2 - 3x - 10 = 0$ has roots α, β, γ .
- i Let $u = -\alpha + \beta + \gamma$. Show that $u + 2\alpha = 1$, and hence find a cubic equation having roots $-\alpha + \beta + \gamma, \alpha - \beta + \gamma, \alpha + \beta - \gamma$.
 - ii State the value of $\alpha\beta\gamma$ and hence find a cubic equation having roots $\frac{1}{\beta\gamma}, \frac{1}{\gamma\alpha}, \frac{1}{\alpha\beta}$.

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4 The roots of the equation

$$x^3 - 8x^2 + 5 = 0$$

are α, β, γ . Show that

$$\alpha^2 = \frac{5}{\beta + \gamma}. \quad [4]$$

It is given that the roots are all real. Without reference to a graph, show that one of the roots is negative and the other two roots are positive. [3]

MJ06

EITHER

Obtain the sum of the squares of the roots of the equation

$$x^4 + 3x^3 + 5x^2 + 12x + 4 = 0. \quad [2]$$

Deduce that this equation does not have more than 2 real roots. [3]

Show that, in fact, the equation has exactly 2 real roots in the interval $-3 < x < 0$. [5]

Denoting these roots by α and β , and the other 2 roots by γ and δ , show that $|\gamma| = |\delta| = \frac{2}{\sqrt{(\alpha\beta)}}$. [4]